admits a simpler analysis. Let each model output a scalar overall score $s_m(x) \in [a_m, b_m]$.

Corollary 2 (Scalar overall-assessment bounds). If each reviewer outputs $s_m(x) \in [a, b]$ and ϕ is the identity, then $\sigma_w^2 = \sum_m w_m^2 \mathrm{Var}[s_m(x)]$ and $c_{\max} = \max_m w_m(b-a)$, hence

$$\Pr\big(\hat{y}(x) \neq y^\star(x)\big) \;\leq\; \exp\left(-\frac{\gamma(x)^2}{2\sigma_w^2 + \frac{2}{3}c_{\max}\gamma(x)}\right). \tag{9}$$

For uniform weights $w_m = 1/M$ and identical per-review variance σ^2, this simplifies to

$$\Pr(\hat{y} \neq y^\star) \leq \exp\left(-\frac{M\,\gamma^2}{2\sigma^2 + \frac{2}{3}(b-a)\gamma}\right), \tag{10}$$

showing that both the variance term σ^2/M and bounded-difference term $(b-a)/M$ scale as $1/M$.

Calibration error and threshold selection. The concentration results above assume a known threshold τ. In practice, however, we must estimate τ from finite calibration data, introducing a second source of error. We now bound this calibration uncertainty. Let $\alpha(\tau) := \mathbb{P}_{x\sim\mathcal{D}_{\mathrm{cal}}}\big[s(x) \geq \tau\big]$ be the true acceptance rate at threshold τ on the calibration distribution, and let $\widehat{\alpha}_{\mathrm{cal}}(\tau)$ be its empirical counterpart (Section 3.4). The calibration set $\{x_i\}_{i=1}^{N_{\mathrm{cal}}}$ is treated as i.i.d. from $\mathcal{D}_{\mathrm{cal}}$.

Proposition 1 (Calibration error bound). For any $\delta \in (0,1)$, with probability at least $1-\delta$ over the draw of $\mathcal{D}_{\mathrm{cal}}$,

$$\sup_{\tau\in\mathbb{R}} \big|\, \widehat{\alpha}_{\mathrm{cal}}(\tau) - \alpha(\tau) \,\big| \;\leq\; \sqrt{\tfrac{1}{2N_{\mathrm{cal}}} \log \tfrac{4}{\delta}}\,. \tag{11}$$

Proof sketch. The class $\{\mathbb{I}[s \geq \tau] : \tau \in \mathbb{R}\}$ has VC dimension 1; apply the Dvoretzky–Kiefer–Wolfowitz (DKW) inequality with VC generalization to obtain (11). □

This uniform bound controls the acceptance-rate error across *all* thresholds simultaneously. For the rate-matching threshold τ_{rate} (defined to match the venue's historical acceptance rate $\alpha^\star$), we therefore have $|\widehat{\alpha}_{\mathrm{cal}}(\tau_{\mathrm{rate}}) - \alpha^\star| \leq \sqrt{\frac{1}{2N_{\mathrm{cal}}} \log \frac{4}{\delta}}$. If $\alpha(\tau)$ is strictly decreasing with slope bounded away from zero near τ_{rate}, this acceptance-rate error translates into a correspondingly small threshold error.

For the probability-consistency threshold $\tau_{0.5}$, the analysis is more delicate because we must estimate the conditional acceptance probability $\pi(t) = \mathbb{P}(y^{\mathrm{hum}} = 1 \mid s(x) \geq t)$ and then invert it. We employ isotonic regression to ensure monotonicity, and the following result bounds the resulting threshold error.

Proposition 2 (Bound for $\tau_{0.5}$ with isotonic calibration). Define the generalized inverses $\tau_{0.5} = \inf\{t : \pi(t) \geq 1/2\}$ and $\hat{\tau}_{0.5} = \inf\{t : \hat{\pi}(t) \geq 1/2\}$. Suppose $\sup_t |\hat{\pi}(t) - \pi(t)| \leq \varepsilon_\pi$ and π has no flat region wider than Δ around $\tau_{0.5}$ and let $c_{\min}$ be the minimal right-slope of π at $\tau_{0.5}$. Then

$$|\,\hat{\tau}_{0.5} - \tau_{0.5}\,| \;\leq\; \min\{\Delta,\, \varepsilon_\pi/c_{\min}\}. \tag{12}$$

Proof sketch. Since π is monotone with right-slope $c_{\min}$, $\pi(\tau_{0.5}+h) \geq \frac{1}{2} + c_{\min}h$ and $\pi(\tau_{0.5}-h) \leq \frac{1}{2} - c_{\min}h$ for $0 < h \leq \Delta$; with $\sup_t |\hat{\pi} - \pi| \leq \varepsilon_\pi$, choosing $h = \min\{\Delta, \varepsilon_\pi/c_{\min}\}$ yields $\hat{\pi}(\tau_{0.5} + h) \geq \frac{1}{2}$ and $\hat{\pi}(\tau_{0.5} - h) \leq \frac{1}{2}$, hence $|\hat{\tau}_{0.5} - \tau_{0.5}| \leq h$. □

Bayesian credible decisions. The frequentist bounds above provide worst-case guarantees but do not directly yield decision rules for individual papers. We complement this analysis with a Bayesian perspective that provides paper-specific uncertainty quantification. Assume $s_m(x) \mid \mu(x) \sim \mathcal{N}(\mu(x), \sigma_m^2)$ independently across m and $\mu(x) \sim \mathcal{N}(\mu_0, \tau_0^2)$. Then the posterior is Gaussian with precision and mean given by

$$\tau_n^{-2} = \tau_0^{-2} + \sum_m \sigma_m^{-2}\,, \tag{13}$$

$$\mu_n = \tau_n^2\left(\mu_0\tau_0^{-2} + \sum_m \sigma_m^{-2}\, s_m\right). \tag{14}$$

For any threshold τ, the posterior decision probability is $\mathbb{P}(\mu(x) \geq \tau \mid \{s_m\}) = 1 - \Phi((\tau - \mu_n)/\tau_n)$. A $1-\alpha$ credible decision is robust (i.e., the credible interval for $\mu(x)$ does not straddle the threshold) whenever $|\tau - \mu_n| \geq z_{1-\alpha/2}\,\tau_n$.

This Bayesian framework also provides a principled rule for soliciting additional reviews. If the current decision is ambiguous ($|\tau - \mu_n| < z_{1-\alpha/2}\,\tau_n$) and a candidate reviewer with variance σ_{new}^2 would resolve the ambiguity in expectation—that is,

$$|\tau - \mu_n| \;\geq\; z_{1-\alpha/2}\,\tau_{n+1}, \quad \tau_{n+1}^{-2} = \tau_n^{-2} + \sigma_{\mathrm{new}}^{-2},$$

then the additional review is worthwhile; otherwise, the expected uncertainty reduction is insufficient to justify the cost. This credible-interval framework thus enables both probability-of-acceptance decisions and adaptive review allocation.

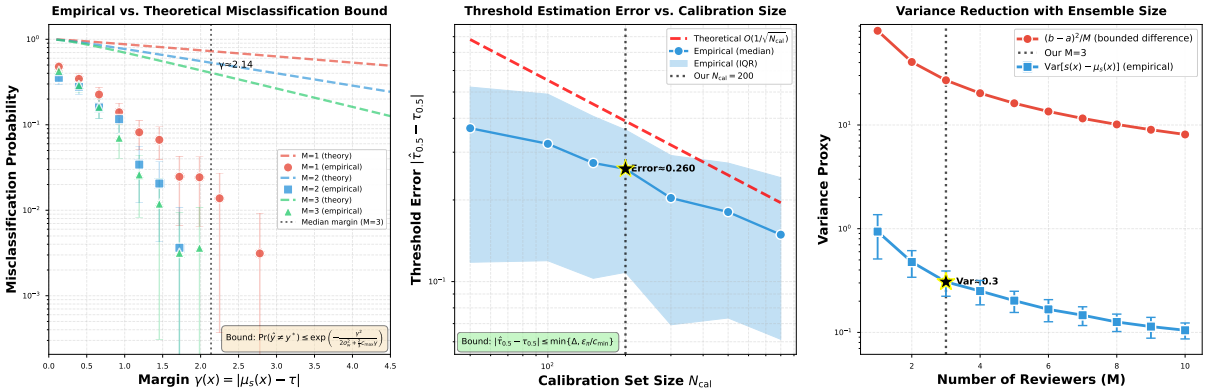

Figure 3: **Empirical validation of error analysis bounds. Left:** Misclassification probability vs. margin $\gamma(x)$ for $M = 1, 2, 3$ reviewers. Empirical rates (points with error bars) fall below theoretical bounds (dashed lines), confirming Eq. (8). **Middle:** Threshold estimation error vs. calibration set size N_{cal}. The blue curve follows the theoretical $O(1/\sqrt{N_{\text{cal}}})$ decay (red dashed); our $N_{\text{cal}} = 200$ (star) yields error ≈ 0.26, validating Proposition 1. **Right:** Variance reduction with ensemble size (log scale). Both the empirical noise variance $\text{Var}[s(x) - \mu_s(x)]$ (blue squares) and the bounded-difference proxy $(b-a)^2/M$ (red circles) decrease as $1/M$, demonstrating that increasing from $M = 1$ to $M = 3$ reviewers reduces both quantities by approximately $3\times$—confirming Theorem 1 and Corollary 2.

Empirical validation. To validate our theoretical bounds, we conduct synthetic experiments that simulate the review aggregation process under controlled conditions. We generate $n = 5{,}000$ synthetic papers with known latent quality scores, each reviewed by $M \in \{1, 2, 3\}$ independent models producing noisy scalar assessments in $[1, 10]$. For each configuration, we compute: (i) empirical misclassification rates as a function of margin $\gamma(x)$ and compare against the bound in (8); (ii) threshold estimation error $|\hat{\tau}_{0.5} - \tau_{0.5}|$ for varying calibration set sizes $N_{\text{cal}} \in \{50, \ldots, 800\}$ via bootstrap with isotonic regression; (iii) empirical noise variance $\text{Var}[s(x) - \mu_s(x)]$ and the bounded-difference proxy $(b-a)^2/M$ as functions of ensemble size M.

Figure 3 presents the results. The left panel confirms that empirical misclassification rates fall well below the theoretical bound across all margins and ensemble sizes, with clear separation between $M = 1, 2, 3$ demonstrating the benefit of aggregation. The middle panel shows threshold error decreasing as $O(1/\sqrt{N_{\text{cal}}})$ as predicted by Proposition 1, with our choice of $N_{\text{cal}} = 200$ (marked by the star) yielding error ≈ 0.26 at the operating point. The right panel demonstrates how increasing the number of reviewers reduces both sources of uncertainty: the empirical noise variance $\text{Var}[s(x) - \mu_s(x)]$ (blue squares) and the bounded-difference proxy $(b-a)^2/M$ (red circles) both decrease as $1/M$. Increasing from $M = 1$ to $M = 3$ reviewers reduces both quantities by approximately $3\times$—confirming that recruiting additional independent reviewers substantially improves decision reliability. These empirical results validate that our bounds correctly characterize the system's behavior.

Practical implications. Taken together, the error analysis in this section yields three actionable recommendations for deploying multi-agent review systems:

(i) Aggregate intelligently. Keep the variance proxy V_w small by recruiting independent reviewers and using variance-aware weighting (e.g., Corollary 1).

(ii) Handle borderline cases carefully. When the margin $|s(x) - \tau|$ is small, use Bayesian credible intervals to assess decision confidence and determine whether additional reviews are needed.

(iii) Calibrate sufficiently. Choose N_{cal} large enough so that the DKW deviation in (11) is negligible at the target confidence level (e.g., $N_{\text{cal}} \geq 200$ for $\delta = 0.05$ yields uniform error $\lesssim 0.11$).

B Seed Topic List

We use `GPT-5` to generate 25 seed topics aligned with the ICLR submission calibration corpus, covering AI, ML, CV, NLP, robotics, systems, and security:

- Self-consistent diffusion models that satisfy counterfactual causal constraints.
- Open-world continual evaluation via synthetic task evolution for multimodal LLMs.
- Mechanistic interpretability of Mixture-of-Experts routing as a cooperative game.
- Certified robustness for retrieval-augmented generation under adversarial knowledge bases.
- Neural field memory: spatially grounded long-horizon memory for vision-language agents.
- Program-of-Thought VLMs with verifiable tool-use and executable intermediate graphs.
- On-device nano-LLMs co-designed with NPU schedulers for sub-1W edge inference.
- Causal video generation: 4D text-to-video with physics-invariant latent constraints.
- Self-curating agents: autonomous dataset construction with legal/ethical compliance proofs.
- Safety proofs for multi-agent LLM protocols under Byzantine participants.
- Open-vocabulary 3D segmentation with Gaussian splats and generative object priors.
- Unlearning at scale: certified removal of concepts from multimodal foundation models.
- Temporal reasoning benchmarks for VLMs built from parametric CAD + differentiable physics.
- Federated reinforcement learning with privacy-preserving credit assignment.
- Energetically aligned training: minimizing carbon under fixed accuracy via differentiable scheduling.
- Watermarking as cryptographic dialogue: interactive proofs to verify AI-generated media.
- Neurosymbolic chart-to-code: parsing scientific plots into executable analysis programs.
- Robust long-form instruction following via adversarial curriculum from self-play reviewers.
- World-model rewrites: editing factual and procedural knowledge in LLMs with locality guarantees.
- Haptic-vision-language models for household manipulation with uncertainty-aware plans.
- Compositional diffusion: plug-and-play constraints for safety, style, and identity preservation.
- Reasoning-first pretraining: supervising latent chains over captions, code, and proofs.
- Open-set alignment: detecting and mitigating unseen harms in generative agents at test time.
- Graph-grounded RAG: joint learning of knowledge graphs and retrievers for verifiable answers.
- RouteBench: measuring strategic routing, tool selection, and delegation in multi-agent LLM systems.